1.60 Large Data Set: OCR MEI (Health)

Worked Solutions

1 a)

17 years old

1 b)

85 years old

1 c)

Divorced, living with partner, married, never married, separated, widowed.

1 d)

Food30 records whether they have eaten within the last 30 minutes

1 e)

Which arm blood pressure is taken on

1 f)

USA

2 a)

kg

2 b)

cm

2 c)

No units

2 d)

cm

2 e)

Beats per second

2 f)

mm Hg

3)

The National Health and Nutrition Examination Survey in the united states.

4)

The systolic blood pressure is the pressure at the time when the heart beats. The diastolic blood pressure is the pressure between heart beats.

5)

Interviewers are instructed to get three measurements of blood pressure (with at least one minute between readings). This might take up to four attempts. They are instructed not to take more than four attempts to get the blood pressure readings.

6)

The average of the 3 readings for systolic and diastolic blood pressure.

7 a)

Simple random sampling

7 b)

There are missing data points in the spreadsheet labelled with a #N/A which might mean that he cannot use that line of data and therefore he ends up with less than 20 pieces of data.

8)

Any person who has a detail missing in the weight, height, BMI, thigh length, upper arm length, waist, Food30, arm, pulse would no longer be available to sample from. Furthermore, she has actively decide to only include people who had their blood pressure readings measured in the first 3 attempts out of 4.

9 a)

She should use stratified sampling.

9 b)

She should remove any lines of data that contain a #N/A in the BMI column.

She would need to find the total number in each of the categories n then use the formula

$$\frac{n}{200 - \text{number of any removed lines}} \times 50$$

to find out how many to sample from the category.

She could then randomly sample the amount she needs from the large data set, using either simple random sampling or systematic sampling, dismissing.

10 a)

Mean is 177.53 cm

Sample standard deviation is 8.81 cm

(standard deviation is 8.36 cm)

10 b)

Whilst there are slightly more males than females, the sample of heights is consistently high except for one shorter male!

10 c)

$$177.53 - 2 \times 8.81 = 159.91 \text{ cm}$$
$$177.53 + 2 \times 8.81 = 195.15 \text{ cm}$$

(if using population:

$$177.53 - 2 \times 8.36 = 160.81 \text{ cm}$$
$$177.53 + 2 \times 8.36 = 194.25 \text{ cm})$$

The male with height 159.4 cm is an outlier.

10 d)

The height is larger on average in the sample than in whole data set.

10 e)

The height is less spread out in the sample than in the whole data set.

11 a)

$$116.3 = 1.95t + 88.70$$
$$t = 14.2 \text{ cm}$$

11 b)

The data in the diagram is close to a straight line, and the PMCC of 0.7518 is strong (positive) correlation.

12 a)

They are both approximately 120 mm Hg

12 b)

Male: ~ 17.5 mm Hg

Female: ~ 30 mm Hg

12 c)

Males and females have the same systolic blood pressure on average, but the systolic blood pressure of males is more consistent than females.

13 a)

Total Frequency

$$= 0.209 \times 10 + 0.179 \times 10 + 0.143 \times 10 + 0.128 \times 10 +$$
$$0.122 \times 10 + 0.097 \times 10 + 0.076 \times 10 + 0.046 \times 10 = 10$$

The median is in the $\frac{10}{2} = 5$th person

$$0.209 \times 10 + 0.179 \times 10 + 0.143 \times 10 = 5.31$$

So the median is in the age 35 to 45 class interval.

13 b)

$\bar{x} = 46$

13 c)

$P(\text{Age} < 45) = 0.209 + 0.179 + 0.143 = 0.531$

So 53.1%

13 d)

There is no upper age limit placed on data collection, so anyone over the age of 95 would be miscategorised. Depending on the age of the oldest person in the LDS, the width of the final class could be reduced or increased (in this case everyone in this class is 85 years old).

14 a)

Let ρ be the population correlation coefficient between BMI and average systolic blood pressure.

$H_0 : \rho = 0$
$H_1 : \rho > 0$
$r_s = 0.8159$

The critical value for $n = 20$ for a one-tailed test at a 5% significance level is 0.3783

$0.8159 > 0.3783$ so we reject H_0

There is sufficient evidence to suggest that there is a positive correlation between average systolic blood pressure.

14 b)

The data on the teacher's graph is distributed across the whole area, which means that it is a closer to zero corelation, so the only way that such a high correlation could have been found is if the data points were picked to be close to a straight line.